15b. Synthetic Difference-in-Differences

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Synthetic Difference-in-Differences* (Causality Lab, Seoul National University, 2025)

The central question is how to combine the strengths of difference-in-differences (invariance to additive unit-level shifts) and synthetic control (re-weighting units to match pre-treatment trends) into one estimator for panel data.

These notes consolidate the slides, write the three estimators as weighted two-way fixed-effects regressions, give the unit- and time-weight problems with their regularization, and compare SDiD with DiD and SCM. Physical PDF page numbers are listed in Section 4. Main reference: Arkhangelsky, Athey, Hirshberg, Imbens and Wager (2021, AER); see also Clarke, Pailañir, Athey and Imbens (2023).

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1 DiD, SC and SDiD

Purpose. Recall DiD and SCM and preview what SDiD changes.

Picture (California Proposition 99, cigarette consumption 1970–2000). *DiD* compares California with the unweighted average of control states, using all pre-periods equally; the level difference is allowed, but the trends must be parallel, and visibly they are not. *SC* re-weights control states so that the synthetic control matches California's level and trend before 1988, using no time weights. *SDiD* re-weights control states so that their trend (not necessarily level) is parallel to California's, and re-weights pre-treatment years, concentrating weight on the few years just before treatment (the shaded time weights).

Recap. DiD estimates effects by comparing outcome changes over time between treated and control groups, under parallel trends (a common trend absent treatment) while allowing time-invariant level differences. SCM addresses comparative case studies with a single treated unit, building a matched synthetic control from many donors. Both are for panel data and similar in spirit, but differ:

(1) SCM adds unit weights;

(2) SCM has time fixed effects but no unit fixed effects and no overall intercept.

SDiD (Arkhangelsky et al. 2021). Like SCM, it re-weights and matches pre-treatment trends, reducing reliance on parallel trends. Like DiD, it is invariant to additive unit-level shifts.

2 Formulations

Purpose. Write all three estimators as (weighted) TWFE regressions.

With unit fixed effects α_i, time fixed effects β_t, an intercept μ and treatment indicator X_{it}, the effect τ is estimated by

$$\hat{\tau}^{\text{DiD}} = \underset{\tau,\mu,\alpha,\beta}{\arg\min} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2, \tag{1}$$

$$\hat{\tau}^{\text{SCM}} = \underset{\tau,\mu,\beta}{\arg\min} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \beta_t - X_{it}\tau)^2\,\hat{\omega}_i, \tag{2}$$

$$\hat{\tau}^{\text{SDiD}} = \underset{\tau,\mu,\alpha,\beta}{\arg\min} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2\,\hat{\omega}_i\hat{\lambda}_t. \tag{3}$$

The unit fixed effects give invariance to additive unit-level shifts; the weights $\hat{\omega}_i$ (and $\hat{\lambda}_t$) match pre-treatment trends.

2.1 Setting and algorithm

A balanced panel with N units and T periods, outcomes Y_{it}, binary exposure $X_{it} \in \{0,1\}$. The first N_{co} units are never exposed; the last $N_{tr} = N - N_{co}$ units are exposed after T_{pre} ($T_{post} = T - T_{pre}$). Like SCM, SDiD finds unit weights $\hat{\omega}$ aligning pre-exposure trends of unexposed units with those of exposed units; it also finds time weights $\hat{\lambda}$ balancing pre-exposure periods with post-exposure ones.

1. Compute unit weights $\hat{\omega}$.
2. Compute time weights $\hat{\lambda}$.

3. Compute $(\hat{\tau}^{\text{SDiD}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i,t}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2 \hat{\omega}_i \hat{\lambda}_t$.

2.2 Unit weights

$$(\hat{\omega}_0, \hat{\omega}) = \underset{\omega_0 \in \mathbb{R},\ \omega \in \Omega}{\arg\min} \left\| \bar{y}_{pre,tr} - (\omega_0 + \omega_{co}^\top Y_{pre,co}) \right\|_2^2 + \zeta^2 T_{pre} \|\omega_{co}\|_2^2,$$

where $\bar{y}_{pre,tr} = \frac{1}{N_{tr}} \sum_{i=N_{co}+1}^{N} Y_{it}$ is the average pre-period outcome path of the treated, $\omega_{co}^\top Y_{pre,co} = \sum_{i=1}^{N_{co}} \omega_i Y_{it}$ is the synthetic control path, and Ω is the simplex (non-negative weights summing to one). Differences from SCM:

1. SDiD allows an intercept ω_0: the synthetic control need not have the same level as the treated units, only a *parallel trend*.
2. The L_2 penalty spreads non-zero weights across control units, instead of the few contributors typical of SCM, forcing more units to be used.

Regularization (optional). $\zeta = (N_{tr} T_{post})^{1/4} \hat{\sigma}(\Delta_{it})$, where $\Delta_{it} = Y_{i(t+1)} - Y_{it}$ is the first difference (temporal fluctuation) of control outcomes in the pre-period,

$$\bar{\Delta} = \frac{1}{N_{co}(T_{pre} - 1)} \sum_{i=1}^{N_{co}} \sum_{t=1}^{T_{pre}-1} \Delta_{it}, \qquad \hat{\sigma}^2(\Delta_{it}) = \frac{1}{N_{co}(T_{pre} - 1)} \sum_{i=1}^{N_{co}} \sum_{t=1}^{T_{pre}-1} (\Delta_{it} - \bar{\Delta})^2,$$

the average fluctuation and the variance of fluctuations.

2.3 Time weights

$\hat{\lambda}$ minimizes the difference between pre- and post-treatment periods for the controls:

$$(\hat{\lambda}_0, \hat{\lambda}) = \underset{\lambda_0 \in \mathbb{R},\ \lambda \in \Lambda}{\arg\min} \left\| \bar{y}_{post,co} - (\lambda_0 + \lambda_{pre}^\top Y_{pre,co}) \right\|_2^2,$$

where $\bar{y}_{post,co} = \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T} Y_{it}$ is a $1 \times N_{co}$ row vector of post-period averages per control unit, $Y_{pre,co}$ is the $T_{pre} \times N_{co}$ matrix of pre-period control outcomes, λ_{pre} a $1 \times T_{pre}$ vector (one entry per pre-period, in the simplex Λ), and λ_0 an intercept allowing the post-period to lie above or below all pre-periods, as in many applications with a clear trend.

3 Comparisons and takeaways

Purpose. Explain what the weights and fixed effects buy.

SDiD versus DiD. DiD solves the same TWFE problem without unit or time weights. The weights make the regression "local": it emphasizes units similar on average to the treated in their past, and periods similar on average to the treated periods. Using only similar units and periods makes the estimator more robust, and the weights can improve precision by implicitly removing systematic, predictable parts of the outcome.

SDiD versus SCM. SCM omits the unit fixed effects and the time weights: $\hat{\tau}^{\text{SCM}} = \arg\min_{\mu,\beta,\tau} \sum_{i,t}(Y_{it} - \mu - \beta_t - X_{it}\tau)^2 \hat{\omega}_i^{\text{SCM}}$. Time weights can remove bias and improve precision by eliminating periods very different from the post-treatment periods. Unit fixed effects make the model more flexible (strengthening robustness) and often explain much of the outcome variation.

Key takeaways.

- Like SCM, SDiD works with multiple periods when pre-treatment trends are not parallel.
- Unlike SCM, its unit weights ω build a control that is only *parallel* to the treated group, not matched in level.
- Like DiD, it uses time (β) and unit (α) fixed effects, which explain outcome variance and reduce the estimator's variance.
- Its unit weights use an L_2 penalty (more spread out across controls) and allow an intercept ω_0.
- It introduces time weights, which neither DiD nor SCM has.

4 Source coverage

Topic	Physical pages
Comparison figure, recap of DiD and SCM, SDiD idea	1–5
Formulations, setting, algorithm	6–8
Unit weights and regularization, time weights	9–12
Comparisons, takeaways, references	13–17